Lecture 10 Rough Stochastic PDEs

Reference: M. Hairer Comm. Pure Appl. Math. 64 (2011), 1547–1585

Consider $du = \partial_x^2 u\,dt + f(u)\,dt + g(u)\partial_x u\,dt + \sigma\,dW_t$ (1)
on $L^2([0,2\pi])$ with periodic bdy conditions, $f, g \in C^3$, and
space-time white noise on L^2, i.e. $W(t,x) = \sum_k \beta_k(t) e_k(x)$

Main goal: give a rigorous meaning to (1), in particular to the "singular" term $g(u)\partial_x u$.

(1) cannot be solved within the variational approach, since $\int g(u)\partial_x u\,u\,dx$ cannot be bounded in terms of $\gamma \|u\|^2_{H^{1,2}} + M\|u\|^2_{L^2}$ so it is not coercive in general.

Going back to the mild approach with $P_t = e^{t\partial_{xx}^2}$ we are looking for a solution of the integral eqn

$$u(t) = P_t u_0 + \int_0^t P_{t-s}\left(f(u_s) + g(u_s)\partial_x u_s\right) ds + \sigma \int_0^t P_{t-s}\,dW_s.$$

Some facts on (P_t): ∂_{xx}^2 has discrete spectrum with eigenvalues $-k^2$ and corresponding eigenfcts.

$$e_0(x) = \frac{1}{\sqrt{2\pi}},\quad e_k(x) = \frac{1}{\sqrt{\pi}}\cos(kx),\quad f_k(x) = \frac{1}{\sqrt{\pi}}\sin(kx),\ k = 1,2,\dots$$

Then
$$\int_0^t P_{t-s}\,dW_s = \underbrace{\int_0^t d\beta_0(s)}_{=\beta_0(t)}\, e_0(x) + \sum_{k=1}^{\infty} \int_0^t e^{-k^2(t-s)}\,d\beta_k(s)\, e_k(x) + \int_0^t e^{-k^2(t-s)}\,d\tilde{\beta}_k(s)\, f_k(x)$$

hence

$$E\left(\left\| \int_0^t p_{t-s}\, dW_s \right\|_{H^{\alpha,2}}^2 \right) = t + 2\sum_{k=1}^{\infty} k^{2\alpha} \underbrace{\int_0^t e^{-2k^2(t-s)}\, ds}_{= \frac{1-e^{-2k^2 t}}{k^2}}$$

$$= t + 2\sum_{k=1}^{\infty} \frac{1-e^{-2k^2 t}}{k^{2(1+\alpha)}} < \infty \quad \text{iff } \alpha < \tfrac{1}{2}$$

Since $\int_0^t p_{t-s}\, dW_s$ has Gaussian distribution this implies, $\int_0^t p_{t-s}\, dW_s \in H^{\alpha,2}$ for all $\alpha < \frac{1}{2}$ but not in $H^{\alpha,2}$ for $\alpha > \frac{1}{2}$.

In particular, no spatial derivative $\partial_x u$ can be expected of the mild solution.

<u>However</u>: Recall the decomposition $u(t,x) = v(t,x) + \sigma W_{\partial_{xx}^2}(t)$, where $W_{\partial_{xx}^2}(t) = \int_0^t p_{t-s}\, dW_s$ and note that we can rewrite

$$v(t,x) = p_t u_0(x) + \int_0^t p_{t-s}\big(f(u(s,\cdot)) + g(u(s,\cdot))\partial_x v(s,\cdot)\big)(x)\, ds$$

$$+ \int_0^t \int_0^{2\pi} p_{t-s}(x-y)\, g(u(s,y)) \underbrace{\partial_x W_{\partial_{xx}^2}(s,y)\, dy}_{= dW_{\partial_{xx}^2}(s,y)}\, ds$$

and for fixed s, the term

$$\int_0^{2\pi} p_{t-s}(x-y)\, g(u(s,y))\, dW_{\partial_{xx}^2}(s,y)$$

is interpreted as a stochastic integral w.r.t. the spatial variable y.
<u>Problem</u>: integrand not adapted to some filtration (both cases $x \le y$, $x \ge y$ occur)!
Therefore, need to extend the notion of Itô-integrals to nonadapted integrands.

A pathwise theory for "stochastic integrals"

Let $X, Y : [0,T] \to \mathbb{R}$ be C^α, $\alpha \in (\frac{1}{3}, \frac{1}{2})$.
We want to define the integral $\int_0^t Y_s \, dX_s$ in terms of Riemann type sums:

$$\int_0^t Y_s \, dX_s = \lim_{|\mathcal{P}| \to 0} \sum_{[s,t] \in \mathcal{P}} Y_s (X_t - X_s).$$

If $Y_t = f(X_t)$, $f \in C^1$, this limit need not exist, but if we expand further

$$\int_s^t Y_r \, dX_r - Y_s (X_t - X_s) = \int_s^t (Y_r - Y_s) \, dX_r = \int_s^t \int_s^r f'(X_u) \, dX_u \, dX_r$$

$$= f'(X_u) \int_s^t (X_r - X_s) \, dX_r + \tilde{R}_{s,t}$$

we can get convergence of

$$\lim_{|\mathcal{P}| \to 0} \sum_{[s,t] \in \mathcal{P}} f(X_s)(X_t - X_s) + f'(X_s) \mathbb{X}_{s,t}$$

provided $\mathbb{X}_{s,t}$, $s \le t$, is $C^{2\alpha}$, satisfies the consistency condition

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = (X_s - u)(X_u - t), \quad s \le u \le t,$$

and $f \in C^2$, and one has the bound

$$\left| \int_s^t f(X_r) \, dX_r - f(X_s)(X_t - X_s) - f'(X_s) \mathbb{X}_{s,t} \right|$$

$$\overset{\shortparallel}{\tilde{R}_{s,t}} \le K \cdot \|f\|_{C_b^2} \left(\|\mathbb{X}\|_{C^{2\alpha}}^3 + \|X\|_{C^\alpha} \|\mathbb{X}\|_{C^{2\alpha}} \right) |t-s|^{3\alpha}$$

and $3\alpha > 1$ implies that $\sum_{[s,t] \in \mathcal{P}} \tilde{R}_{s,t} \to 0$, $|\mathcal{P}| \to 0$.

Stationary solutions

Consider $\quad dX_t = (\partial_{xx}^2 - 1) X_t \, dt + \sigma \, dW_t \qquad (*)$

where the semigroup

$$S_t = e^{-t} P_t \text{ of } (\partial_{xx}^2 - 1) \text{ is exponentially decreasing}$$

It follows that

$$X_t = \sigma \int_{-\infty}^{t} e^{-(t-s)} P_{t-s} \, dW_s \quad , \; t \in \mathbb{R},$$

is a stationary solution to $(*)$, with (stationary) distribution

$$\mathcal{N}\Big(0, \sigma^2 \underbrace{\int_0^\infty e^{-2t} P_{2t} \, dt}_{= \frac{1}{2}(Id - \partial_{xx}^2)^{-1} : L^2([0,2\pi]) \to L^2([0,2\pi])}\Big).$$

Note: $\sigma^2 \int_0^\infty e^{-2t} P_{2t} f \, dt = \sigma^2 \int_0^\infty e^{-2t} dt \, \langle f, e_0 \rangle e_0 + \sum_{k=1}^\infty \sigma^2 \int_0^\infty e^{-2(1+k^2)t} dt \big(\langle f, e_{2k} \rangle e_{2k} + \langle f, e_{2k+1} \rangle e_{2k+1} \big)$

$$= \frac{\sigma^2}{2} \langle f, e_0 \rangle e_0 + \sum_{k=1}^\infty \frac{\sigma^2}{2(1+k^2)} \big(\langle f, e_{2k} \rangle e_{2k} + \langle f, e_{2k+1} \rangle e_{2k+1} \big)$$

$$= \int f(y) K(\cdot - y) \, dy \, ,$$

where

$$K(x) = \frac{\sigma^2}{2\pi} \sum_{k \in \mathbb{Z}} \frac{\cos(kx)}{1+k^2} = \frac{\sigma^2}{2} \frac{\cosh(|x| - \pi)}{\sinh(\pi)} \, , \quad x \in [-\pi, \pi] \text{ with periodic extension}$$

Then X_t for fixed t can be lifted to a rough path $(X_t, \mathbb{X}_t)$, where

$$\mathbb{X}_t(x,y) = \lim_{\varepsilon \to 0} \int_x^y (X_t^\varepsilon(v) - X_t^\varepsilon(x)) \, \partial_x X_t^\varepsilon(v) \, dv \quad \text{a.s.}, \; 0 \le x \le y \le 2$$

for any (approximating) sequence $(X_t^\varepsilon(x))_\varepsilon$ of Gaussian processes

smooth w.r.t. x and such that

$$\sup_x E\left(|X_t^\varepsilon(x) - X_t(x)|^2\right) \to 0 .$$

Moreover, for every $\alpha < \frac{1}{2}$, there exists $\theta > 0$ and C_q s.th.,

$$E\left(\|X_s - X_t\|_\alpha^{2q} + \|\mathbb{X}_s - \mathbb{X}_t\|_{2\alpha}^q\right) \le C_q |t-s|^{\theta q} \qquad \forall q \ge 1$$

and every $s, t \in [0,T]$.

Here, $\|u\|_\alpha = \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x-y|^\alpha}$, $\|\mathbb{U}\|_{2\alpha} = \sup_{x \le y} \frac{|\mathbb{U}(x,y)|}{|x-y|^{2\alpha}}$.

Consider now the integral equation

$$v(t,x) = e^{-t} P_t(u_0 - X_0)(x) + \int_0^t e^{-(t-s)} P_{t-s}\left(g(u(s,\cdot)) \partial_x v(s,\cdot) + \tilde{f}(u(s,\cdot))\right)(x)$$

(2)

$$+ \int_0^t \int_0^{2\pi} e^{-(t-s)} p_{t-s}(x-y)\, g(u(s,y))\, dX_s(y)\, ds , \quad \tilde{f}(u) = f(u) + u$$

<u>Theorem:</u> Let $\beta \in (\frac{1}{3}, \frac{1}{2})$ and $u_0 \in C^\beta$. Then for a.e. realization of X, there exists $T > 0$ s.th. (1) has a unique mild solution taking values in $C([0,T]; C^\beta)$. If g is bdd and all derivatives of f and g are bdd, then this solution is global (i.e., T independent of X).

Proof: Picard iteration for $v(t,x) = u(t,x) - X_t(x) - \overbrace{e^{-t}P_t(u_0 - X_0)(x)}^{=:\mathcal{U}_t(x)}$ satisfies the following modification of (2):

$$v(t,x) = \int_0^t e^{-(t-s)} P_{t-s}\Big(g(u(s,\cdot))\big(\partial_x v(s,\cdot) + \partial_x \mathcal{U}_s(\cdot)\big) + \hat{f}(u(s,\cdot))\Big)(x)\,ds$$

$$+ \int_0^t \int_0^{2\pi} e^{-(t-s)} p_{t-s}(x-y)\, g(u(s,y))\, dX_s(y)\, ds$$

$$=: \mathcal{M}^{(1)}_{T,X} v(t,x) + \mathcal{M}^{(2)}_{T,X} v(t,x), \quad T \le 1 \text{ to be def. later.}$$

<u>Step 1:</u> $\mathcal{M}^{(1)}_{T,X}$ (classical)

(i) $\|\mathcal{M}^{(1)}_{T,X} v\|_{1,T} = \sup_{t\le T} \|\mathcal{M}^{(1)}_{T,X} v(t)\|_{C^1} \le C_K(1 + |||X|||)$,

for $\|u_0\|_{C^\beta} \le K$, $\|X_0\|_{C^\beta} \le K$, $\|v\|_{1,T} \le K$.

Here, $|||X||| := \sup_{t\le 1}\big(\|X_t\|_{C^\alpha} + \|X_t\|_{C^{2\alpha}}\big)$ for a fixed $\alpha \in (\frac{1}{3}, \beta)$.
Main ingredient: $\|\mathcal{U}_s\|_{C^1} \le c K s^{\frac{\beta-1}{2}}$ and $\|P_s \tilde{v}\|_{C^1} \le K \|\tilde{v}\|_\infty$.

(ii) Modulus of continuity

$$(\mathcal{M}^{(1)}_{T,X} v - \mathcal{M}^{(1)}_{T,X} \bar{v})(t) = \int_0^t e^{-(t-s)} P_{t-s}\big(g(u_s)(\partial_x(v_s - \bar{v}_s) + \hat{f}(u_s) - \hat{f}(\bar{u}_s))\big)\,ds$$

$$+ \int_0^t e^{-(t-s)} P_{t-s}\big((g(u_s) - g(\bar{u}_s))(\partial_x \bar{v}_s + \partial_x \mathcal{U}_s)\big)\,ds$$

implies

$$\|\mathcal{M}^{(1)}_{T,X} v - \mathcal{M}^{(1)}_{T,X} \bar{v}\|_{1,T} \le C_K(1 + |||X|||)\, \|v - \bar{v}\|_{1,T}\, T^{\beta/2}$$

if g, g', f' are bounded.

Step 2 $\mathcal{M}^{(2)}_{T,X}$ (new)

Main ingredients: let $\alpha \in (\frac{1}{3}, \frac{1}{2})$, then

$$(*) \quad \left| \partial_x \int_0^{2\pi} e^{-k} p_t(x-y) Y_y \, dX(y) \right| \leq C t^{\frac{\alpha}{2}-1} \|Y\|_{X,\alpha} \, |||\mathbb{X}||| \text{ uniformly on } [0,2\pi].$$

using

$$\partial_x p_t(x) = -\sum_{n \in \mathbb{Z}} \frac{x}{\sqrt{2\pi}\, t^{3/2}} e^{-\frac{(x-2\pi n)^2}{2t}} = \frac{1}{t} f_t\left(\frac{x}{\sqrt{t}}\right)$$

for some fct f_t satisfying $\sup_{t \in (0,1]} \|f_t\|_{1,1} < \infty$.

The estimate $(*)$ is the crucial estimate and follows from:

$$\int_0^{2\pi} Z_y \, dX(y) = \lim_{|\mathcal{P}| \to 0} \sum_{[x,y] \in \mathcal{P}} \left(Z_x (X(y) - X(x)) + Z'_x \mathbb{X}(x,y) \right)$$

where Z'_x is such that the remainder

$$(*) \quad R_{x,y} = Z_y - Z_x - Z'_x (X_y - X_x) \text{ satisfies } \|R\|_{2\alpha} < \infty,$$

"Gubinelli derivative" (not uniquely specified)

which implies that

$$(Z, Z') \longmapsto \left(\int_0 Z_y \, dX_y, \; Z \right) \text{ is continuous}$$

on $C^\alpha_{\mathbb{X}} := \{ (Z, Z') : (*) \text{ holds} \}$ with norm

$$\|Z\|_{X,\alpha} := \|Z\|_{C^\alpha} + \|Z'\|_{C^\alpha} + \|R\|_{2\alpha}$$

More specifically,

$$\left\| \int_0 (Z_y - Z_0) \, dX_y \right\|_\alpha \leq C \left(\|X\|_\alpha \|R\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Z'\|_\alpha \right).$$

This is applied to $Y_x = g(v(s,x) + X_s(x) + u_s(x))$

$$Y'_x = g'(v(s,x) + X_s(x) + u_s(x)),$$

which yields

$$\|Y\|_{X,\alpha} \le C_K \left(\|v_s\|_{C^{2\alpha}} + \|u_s\|_{C^{2\alpha}}\right)\left(1+|||X|||\right)^2$$

(cf lemma 2.2 in Hairer, op cit.) $\quad \le C_K \left(1+s^{+\left(\frac{\beta-2\alpha}{2}\right)}\right)\left(1+|||X|||\right)^2.$

Summarizing:

$$\left|\partial_x \int_0^{2\pi} p_{t-s}(x-y)\, Y_y \, dX(y)\right| \le C_K \left(1+|||X|||\right)^3 \left(1+s^{\frac{\beta-2\alpha}{2}}\right)(t-s)^{\frac{\alpha}{2}-1}$$

and thus

$$\|\mathcal{M}^{(2)}_{T,X} v\|_{1,T} \le C_K \left(1+|||X|||\right)^3 \underbrace{\sup_t \int_0^t \left(1+s^{\frac{\beta-2\alpha}{2}}\right)(t-s)^{\frac{\alpha}{2}-1} ds}_{\le C T^{\frac{\beta-\alpha}{2}}}.$$

For the modulus of continuity:

$$\mathcal{M}^{(2)}_{T,X} v(t) - \mathcal{M}^{(2)}_{T,X} \bar{v}(t) = \int_0^t \int_0^{2\pi} e^{-t} p_{t-s}(\cdot - y) \Big[g(v(s,y) + X_s(y) + u_s(y)) - g(\bar{v}(s,y) + X_s(y) + u_s(y)) \Big] dX(y)\, ds$$

$$= \int_0^t \int_0^{2\pi} e^{-t} p_{t-s}(\cdot - y) \Big[\int_0^1 g'(X_s(y) + u_s(y) + v_\lambda(s,y)) \left(v(s,y) - \bar{v}(s,y)\right) d\lambda \Big] dX_s(y)\, ds$$

and use that

$$\left\| \int_0^1 g'(\dots)(v(s,y) - \bar{v}(s,y))\, d\lambda \right\|_{X_s,\alpha} \le C_K \left(1+|||X|||\right)^2 \left(1+s^{\frac{\beta-2\alpha}{2}}\right) \cdot \|v_s - \bar{v}_s\|_{C^{2\alpha}}$$

We therefore obtain

$$\|\mathcal{M}^{(2)}_{T,X} v - \mathcal{M}^{(2)}_{T,X} \bar{v}\|_{1,T} \leq C_K (1 + |||X|||)^3 T^{\frac{3-\alpha}{2}} \|v - \bar{v}\|_{1,T}.$$

We arrive at a contraction for small T, hence a local in time solution to (2).

If g, g', g'', f' are bounded, then T^* can be chosen independent of X s.th. $\|\mathcal{M}_{T,X} v\|_{1,T} \leq \frac{K}{2}\|v\|$ for all $T \leq T^*$. It follows that the local solution can be extended to a global solution. ∎

<u>Stability of the solution</u>

We have found a "good" fixed point representation of the solution u of (1) in this sense that it can be realized as the limit of smooth noise approximations:

<u>Corollary</u>: Let $\varphi \in C_0(\mathbb{R})$, $\int \varphi(x)dx = 1$, $\varphi_\varepsilon(x) = \frac{1}{\varepsilon}\varphi(\frac{x}{\varepsilon})$, hence $\varphi_\varepsilon(x)dx \longrightarrow \delta_0(dx)$. Let $Q_\varepsilon u(x) := \int_{S^1} \varphi_\varepsilon(x-y)u(y)dy$.
Then

$$du_\varepsilon = \partial_x^2 u_\varepsilon dt + f(u_\varepsilon)dt + g(u_\varepsilon)\partial_x u_\varepsilon dt + \sigma Q_\varepsilon dW_t$$

has a unique mild solution (up to some blow up time τ_ε)

$$u_\varepsilon(t,x) = v_\varepsilon(t,x) + \sigma W^\varepsilon_{\partial_{xx}^2}(t), \quad W^\varepsilon_{\partial_{xx}^2}(t) = \int_0 P_{t-s} Q_\varepsilon dW_s$$

where v_ε solves

$$v_\varepsilon(t,x) = P_t u_0(x) + \int_0^t P_{t-s}\big(f(u_\varepsilon(s,\cdot)) + g(u_\varepsilon(s,\cdot))\partial_x v_\varepsilon(s,\cdot)\big)(x)ds$$
$$+ \int_0^t \int_0^{2\pi} P_{t-s}(x-y)\, g(u_\varepsilon(s,y))\, \partial_{xx}^2 W^\varepsilon(s,y)\, dy\, ds$$

This time, $\partial_x \omega^{\varepsilon}_{\partial_{xx}^2}(s) \in L^2$ with $E(\|\partial_x \omega^{\varepsilon}_{\partial_{xx}^2}(s)\|^2_{L^2}) \leq \frac{2\pi}{\varepsilon^3}\int \mathcal{C}'(x)^2\,dx$

hence the rough integral $\int_0^{2\pi} p_{t-s}(x-y)\, g(u_\varepsilon(s,y))(\partial_x W^{\varepsilon}_{\partial_{xx}^2})(s,y)\,dy$ exists in the classical sense. Then there exist $\hat{\tau}_\varepsilon \leq \tau_\varepsilon$ converging to τ a.s. such that $\lim_{\varepsilon\to 0} \sup_{t\leq \hat{\tau}_\varepsilon} \|u_\varepsilon(t) - u(t)\|_{C^\beta} = 0$ ∎

The proof follows from the general stability result below and the fact that the (continuous version of)

$$X^{\varepsilon}_t = \sigma \int_{-\infty}^{t} e^{-(t-s)} P_{t-s}\, Q_\varepsilon\, dW_s\,, \quad t \in \mathbb{R}$$

with $$\mathbb{X}^{\varepsilon}_t(x,y) = \int_x^y (X^{\varepsilon}_t(v) - X^{\varepsilon}_t(x))\, \partial_x X^{\varepsilon}_t(v)\,dv$$

converges to $(X_t, \mathbb{X}_t)$ uniformly in the rough path metric. (see details in Hairer, op. cit.)

<u>Theorem (Stability result)</u>: Let $u_0, \bar{u}_0 \in C^\beta$, $\mathbb{X}, \bar{\mathbb{X}} \in C([0,T]; \mathcal{D}^\alpha)$ (= space of rough paths with norm $|||\mathbb{X}||| = \sup \|X\|_{C^\alpha} + |||\mathbb{X}|||_{C^{2\alpha}}$), with $\frac{1}{3} < \alpha < \beta < \frac{1}{2}$. Let $u, \bar{u}$ be the corresponding local solutions with blow up time $\tau, \bar{\tau}$. Then for every $T < \tau$ and every $\varepsilon > 0$ there exists $\delta > 0$ for which $\bar{\tau} \geq T$ and

$$\sup_{t\leq T} \|u_t - \bar{u}_t\|_{C^\beta} \leq \varepsilon$$

for all $\bar{u}_0, \bar{\mathbb{X}}$ such that $\|u_0 - \bar{u}_0\|_{C^\beta} + \sup_{t\leq T}\|X_t - \bar{X}_t\|_{C^\beta} + \sup_{t\leq T}\|\mathbb{X}_t - \bar{\mathbb{X}}_t\|_{C^{2\alpha}} \leq \delta$. ∎

<u>Remark:</u> $\partial_x W^{\varepsilon}_{\partial^2_{xx}}(s) \in L^2$ with $E\left(\|\partial_x W^{\varepsilon}_{\partial^2_{xx}}(t)\|^2_{L^2}\right) \leq \frac{2\pi}{\varepsilon^3}\int \varphi'(x)^2\,dx$

follows from

$$\partial_x(Q_\varepsilon u)(x) = \frac{1}{\varepsilon^2}\int \varphi'\left(\frac{x-y}{\varepsilon}\right) u(y)\,dy =: T_\varepsilon u(x).$$

Hence $\partial_x\left(\int_0^t p_{t-s} Q_\varepsilon\, dW_s\right) = \beta_0(t)\, T_\varepsilon e_0 + \sum_{k=1}^{\infty} \int_0^t e^{-k^2(t-s)}\, d\beta_k(s)\, T_\varepsilon e_k$
$+ \int_0^t e^{-k^2(t-s)}\, d\tilde{\beta}_k(s)\, T_\varepsilon f_k$)

satisfies

$$E\left(\|\partial_x\left(\int_0^t p_{t-s} Q_\varepsilon\, dW_s\right)\|^2_{L^2}\right) \underset{\substack{\text{Ind. of}\\ \beta_k,\tilde{\beta}_k}}{\leq} \|T_\varepsilon e_0\|^2 + \sum_{k=1}^{\infty} \frac{1-e^{-k^2 t}}{k^2}\left(\|T_\varepsilon e_k\|^2 + \|T_\varepsilon f_k\|^2\right)$$

$$\leq \|T_\varepsilon\|^2_{L_2(L^2)} = \frac{2\pi}{\varepsilon^2}\int \varphi'(x)^2\,dx$$

Hilbert-Schmidt

Since

$$\|T_\varepsilon\|^2_{L_2(L^2)} = \sum_k \|T_\varepsilon g_k\|^2_{L^2} = \frac{1}{\varepsilon^4}\sum_k \int_{S^1}\left(\int_{S^1} \varphi'\left(\frac{x-y}{\varepsilon}\right) g_k(y)\,dy\right)^2 dx$$

$$= \frac{1}{\varepsilon^4}\int_{S^1} \underbrace{\left\|\varphi'\left(\frac{x-\cdot}{\varepsilon}\right)\right\|^2_{L^2(S^1)}}_{\leq \varepsilon\int_{\mathbb{R}} \varphi'(x)^2\,dx}\,dx \leq \frac{2\pi}{\varepsilon^3}\int \varphi'(x)^2\,dx \quad \blacksquare$$

Appendix: Continuity of the heat semigroup in Hölder-spaces

$$p_t(x,y) = \frac{1}{(2\pi t)^{n/2}} e^{-\frac{\|x-y\|^2}{2t}} \quad \text{on } \mathbb{R}^n, \ t>0.$$

Lemma: (i) $[p_t v]_{C^\beta} \le [v]_{C^\beta}$ for all t, β,

(ii) $\|\nabla p_t v\|_\infty \le C_n \, t^{\frac{\beta-1}{2}} [v]_{C^\beta}$ for all t, β.

Proof: (i) $$|p_t v(x) - p_t v(y)| = \frac{1}{(2\pi t)^{n/2}} \left| \int_{\mathbb{R}^n} v(x+u) - v(y+u) \, e^{-\frac{\|u\|^2}{2t}} \, du \right|$$

$$\le [v]_{C^\beta} \|x-y\|^\beta.$$

(ii) $$\nabla p_t v(x) = \frac{1}{(2\pi t)^{n/2}} \int v(u) \frac{u-x}{t} e^{-\frac{\|x-u\|^2}{2t}} \, du$$

$$= \frac{1}{(2\pi t)^{n/2}} \int (v(u) - v(x)) \frac{(u-x)}{t} e^{-\frac{\|x-u\|^2}{2t}} \, du$$

$$\Rightarrow \|\nabla p_t v(x)\| \le \frac{1}{(2\pi t)^{n/2}} \int \frac{\|u-x\|^{1+\beta}}{t} e^{-\frac{\|x-u\|^2}{2t}} [v]_{C^\beta}$$

$$\le C_n \, t^{\frac{1+\beta}{2}-1} = C_n \, t^{\frac{\beta-1}{2}} \quad \blacksquare$$